Low Poly Dungeon FREE

Modular dungeon kit - documentation and setup guide

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1. What this pack is

200 low-poly dungeon models, built to snap together on one 4.00 m grid. Modular walls, floors, ceilings and stairs; doors and gates; torches and braziers; crates, cages and the dressing that makes a dungeon look lived in.

Everything is authored in metres at real-world scale, so a model dropped into a scene is the size the thing would be.

Models	200
Triangles	312 to 3,944, averaging 1,850
Materials	3 - opaque, glass, emissive
Textures	1024x1024 colour atlas, plus a metallic/smoothness atlas in the same layout, and masked glass and emissive maps
Format	FBX

2. Importing

Import the package. Everything lands under Assets/LowPolyDungeonFree.

The models arrive already textured: each FBX names the pack's materials, and its importer settings bind them by name.

If the project uses URP or HDRP, the pack converts its materials on first import. If anything looks magenta, run the convert menu item below.

Then run **Tools > LowPolyDungeonFree > Build Prefabs** once.

3. The grid

The kit's one promise is that the pieces meet. Every join in the demo scenes is measured tight to within a tenth of a millimetre.

Everything drops at y = 0 - floors, walls, stairs alike. A wall's bottom 0.20 m is buried in the floor slab, which is what lets one placement height work for every piece and gives vertical tolerance on uneven floors.

Tile	4.00 x 4.00 m
Wall piece	4.00 m long, 0.40 m thick, 4.00 m tall
Floor slab	4.00 x 4.00 m, 0.20 m thick
Clear height	3.80 m, floor surface to ceiling
Doorway	1.60 x 2.60 m
Arch	2.80 m span, springing at 1.90 m

Walls are dressed on BOTH faces, so one piece serves two rooms rather than two pieces back to back.

4. Pivots and orientation

Every model is pivoted at the bottom-centre of its bounds, so dropping it at y = 0 stands it on the floor.

Doors, gates, cell doors and trapdoors are pivoted on their HINGE EDGE instead, so rotating the prefab swings the leaf on its hinges rather than spinning it about its middle.

Every model faces Unity +Z with no rotation on its root object, so transform.forward points out of the front of a chest or a door.

5. Materials and recolouring

The pack is three materials and one real texture. Every face is UV-mapped to the centre of one cell in the colour atlas, and the metallic/smoothness atlas uses the same cell layout, so iron reads as metal and oak does not.

Recolour the whole kit by editing LPD_Palette.png. Every model that used a swatch changes with it. Keep point filtering on: a face sits at a cell CENTRE, and bilinear filtering across a cell edge would blend two colours along every seam.

The emissive material is what makes a lit torch glow. Lit and unlit versions of torches, braziers, sconces and fire pits are separate models, identical apart from the fire, so a room can be lit by swapping a prefab.

6. Fire, and what moves

THE FIRE IS ANIMATED. Every lit model in the pack - torches, sconces, cressets, braziers, fire pits, candles, candelabra, lanterns, hanging lamps, chandeliers, censers, hearths, forges, coal beds and the flame-vent trap - gets a flame particle system and a flickering light when you run Build Prefabs. Nothing is baked into the mesh; the models themselves are static geometry, which is what keeps them cheap.

FX_LPD_Fire.prefab is the particle effect: a cone emitter with a colour ramp from pale flame through ember to smoke, rendered on the pack's own emissive material so it needs no extra texture and adds no draw call beyond the particle one. It is parented to every lit prefab automatically, positioned and sized from the bounds of that model's OWN emissive faces - so a candle gets a candle's flame and a forge gets a forge's.

LPD_Flicker.cs is the light: it ships in Runtime/, not Editor/, so it compiles into builds and a buyer's torches do not go dead-constant the moment they press Play. Speed and depth are per-instance, and it drives intensity and range together the way a real flame does.

Fire is detected BY MATERIAL, not by name. A model burns if it carries one of the palette's fire swatches on its emissive faces, which is why every candle, lantern, censer and coal bed is covered - name matching found ten of the fifty-one.

The arcane emissives - summoning circles, runestones, glowing crystal and glowing fungus - deliberately do NOT flicker. They glow steadily, because a rune that gutters like a candle reads as a broken light rather than as magic.

Unlit twins of every fixture ship alongside the lit ones, identical apart from the fire, so a dark corridor costs nothing.

7. Editor menu reference

All of these live under **Tools > LowPolyDungeonFree**.

7.1 Build Prefabs

Walks the Models folder and writes one prefab per FBX, each carrying a mesh renderer, a mesh collider and the pack's shared materials. Run it once after import; it is safe to run again.

7.2 Build Library Scene

Lays every model out on a grid in a new scene, so the whole kit can be looked through without dragging prefabs one at a time.

7.3 Convert Materials to This Pipeline

Detects the active render pipeline and converts the pack's three materials to it. It runs by itself on first import and decides per material by reading that material's shader, so importing into a second URP or HDRP project converts correctly too.

8. Render pipelines

Built-in, URP and HDRP are supported. The pack ships on the Built-in Standard shader and converts on import.

The metallic/smoothness atlas is carried across under each pipeline's own property names: _MetallicGlossMap on Built-in and URP, and _MaskMap on HDRP, where its green channel carries ambient occlusion.

9. Performance

Models run 312 to 3,944 triangles, averaging 1,850. The heaviest pieces are the hero props; the modular walls and floors, which repeat most, are deliberately the cheapest.

Three materials means three draw calls for a fully dressed room before batching. Enable GPU instancing on the materials for large levels: the kit is built from repeated pieces, which is the case instancing is for.

10. Support

If something does not behave as this guide says, reply on the Asset Store page for this pack with your Unity version and render pipeline.

The complete kit - every category here plus crypt, arcane, armoury, treasure, traps and cave dressing - is on the same publisher page.